

Session 11 task

B1:

Answer: Does NOT compile.

Reason:

Shape is an abstract class, and abstract classes cannot be instantiated directly.

Code:

```
Shape s = new Shape();
```

Result: Compile-time error.

B2:

Answer: Does NOT compile.

Reason:

Square inherits from Shape but does not implement the abstract method GetArea(). Therefore, Square must either override GetArea() or be declared abstract.

Result: Compile-time error.

B3:

Answer: Compiles successfully.

Output:

Meow

Reason:

Animal reference points to a Cat object. Because MakeSound() is virtual and overridden in Cat, dynamic dispatch calls Cat.MakeSound().

7. Print everyone (List<Person> - Abstraction)

```
Choose an option:
7
Person: youssef
  Year 3, GPA 2.00
  Role: Student
Person: khaled
  3 years of experience
  Role: Instructor
Person: osama
  Role: Admin

=== StudentPortal Menu (Session 11 Capstone) ===
1. Register a new student
2. Register a new instructor
3. Register a new Admin
4. Create a course
5. Enroll a student in a course
6. Compare two students by GPA (>)
7. Print everyone (List<Person> - Abstraction)
8. Print everything printable (List<IPrintable> - Interfaces)
9. Show total students/courses ever created
10. print everything
11. GetRankScore
0. Quit
Choose an option:
```

8. Print everything printable (List<IPrintable> - Interfaces)

```
Choose an option:
8
Person: youssef
  Year 2, GPA 2.00
Person: khaled
  3 years of experience
Person: khaled
  3 years of experience
Person: osama
Access Level : 2
```

10. print everything

```
0. Quit
Choose an option:
10
Person: youssef
    Year 2, GPA 2.00
Person: khaled
    3 years of experience
Person: khaled
    3 years of experience
Person: osama
Access Level : 2
```

11. GetRankScore

```
0. Quit
Choose an option:
11
accesslevel rank : 2
```

